

5 Dedekind extensions

In this lecture we prove that the integral closure of a Dedekind domain in a finite extension of its fraction field is also a Dedekind domain; this implies, in particular, that the ring of integers of a number field is a Dedekind domain. We then consider the factorization of prime ideals in Dedekind extensions.

5.1 Norms and traces

We now introduce the norm and trace map associated to a finite free ring extension B/A . These are often defined only for field extensions, but in fact the same definition works without modification whenever B is a free A -module of finite rank. One can generalize further to projective modules (with some restrictions), but we will not need this.

Definition 5.1. Let B/A be a (commutative) ring extension in which B is a free A -module of finite rank. The (relative) *norm* $N_{B/A}(b)$ and *trace* $T_{B/A}(b)$ of b (down to A) are the determinant and trace of the A -linear multiplication-by- b map $B \rightarrow B$ defined by $x \mapsto bx$.

As a special case, note that if A is a field and B is a finite A -algebra (a field extension, for example) then B is an A -vector space of finite dimension, hence a free A -module of finite rank. In practice one computes the norm and trace by picking a basis for B as an A -module and computing the matrix of the multiplication-by- b map with respect to this basis; this is an $n \times n$ matrix with entries in A whose determinant and trace are basis independent.

It follows immediately from the definition that $N_{B/A}$ is multiplicative, $T_{B/A}$ is additive, we have group homomorphisms

$$N_{B/A}: B^\times \rightarrow A^\times \quad \text{and} \quad T_{B/A}: B \rightarrow A,$$

and if B_1/A and B_2/A are two ring extensions that are free A -modules of finite rank then

$$N_{B_1 \times B_2/A}(x) = N_{B_1/A}(x_1)N_{B_2/A}(x_2) \quad \text{and} \quad T_{B_1 \times B_2/A}(x) = T_{B_1/A}(x_1) + T_{B_2/A}(x_2)$$

for all $x = (x_1, x_2) \in B_1 \times B_2$.

Example 5.2. Consider $A = \mathbb{R}$ and $B = \mathbb{C}$, which has the A -module basis $(1, i)$. For $b = 2 + 3i$ the matrix of $B \xrightarrow{\times b} B$ with respect to this basis can be written as $\begin{pmatrix} 2 & -3 \\ 3 & 2 \end{pmatrix}$, thus

$$N_{\mathbb{C}/\mathbb{R}}(2 + 3i) = \det \begin{pmatrix} 2 & -3 \\ 3 & 2 \end{pmatrix} = 13,$$

$$T_{\mathbb{C}/\mathbb{R}}(2 + 3i) = \text{tr} \begin{pmatrix} 2 & -3 \\ 3 & 2 \end{pmatrix} = 4.$$

Warning 5.3. In order to write down the matrix of an A -linear transformation $B \rightarrow B$ with respect to basis for B as a free A -module of rank n , we not only need to pick a basis, we need to decide whether to represent elements of $B \simeq A^n$ as row vectors with linear transformations acting via matrix multiplication on the right, or as column vectors with linear transformations acting via matrix multiplication on the left. The latter convention is often implicitly assumed in the literature (as in the example above), but the former is often used in computer algebra systems (such as Magma).

We now verify that the norm and trace are well behaved under base change.

Lemma 5.4. Let B/A be ring extension with B free of rank n over A , and let $\varphi: A \rightarrow A'$ be a ring homomorphism. The base change $B' = B \otimes_A A'$ of B to A' is a free A' -module of rank n , and for every $b \in B$ we have

$$\varphi(N_{B/A}(b)) = N_{B'/A'}(b \otimes 1) \quad \text{and} \quad \varphi(T_{B/A}(b)) = T_{B'/A'}(b \otimes 1).$$

Proof. Let $b \in B$, let $(b_1, \dots, b_n)$ be a basis for B as an A -module, and let $M = (m_{ij}) \in A^{n \times n}$ be the matrix of $B \xrightarrow{\times b} B$ with respect to this basis. Then $(b_1 \otimes 1, \dots, b_n \otimes 1)$ is a basis for B' as an A' -module (thus B' is free of rank n over A') and $M' = (\varphi(m_{ij})) \in A'^{n \times n}$ is the matrix of $B' \xrightarrow{\times b \otimes 1} B'$, and we have

$$\begin{aligned} \varphi(N_{B/A}(b)) &= \varphi(\det M) = \det M' = N_{B'/A'}(b \otimes 1) \\ \varphi(T_{B/A}(b)) &= \varphi(\text{tr } M) = \text{tr } M' = T_{B'/A'}(b \otimes 1) \end{aligned} \quad \square$$

Theorem 5.5. Let K be a field with separable closure Ω and let L be an étale K -algebra. For all $\alpha \in L$ we have

$$N_{L/K}(\alpha) = \prod_{\sigma \in \text{Hom}_K(L, \Omega)} \sigma(\alpha) \quad \text{and} \quad T_{L/K}(\alpha) = \sum_{\sigma \in \text{Hom}_K(L, \Omega)} \sigma(\alpha).$$

Proof. Let n be the rank of L as a K -module. By the previous lemma and Proposition 4.42,

$$N_{L/K}(\alpha) = N_{(L \otimes_K \Omega)/\Omega}(\alpha \otimes 1) = N_{\Omega^n/\Omega}(\sigma_1(\alpha), \dots, \sigma_n(\alpha)) = \prod_{i=1}^n \sigma_i(\alpha).$$

The isomorphism $L \otimes_K \Omega \rightarrow \prod_{\sigma} \Omega = \Omega^n$ of Prop. 4.42 sends $\alpha \otimes 1$ to $(\sigma_1(\alpha), \dots, \sigma_n(\alpha))$. Using the standard basis for Ω^n , the matrix of multiplication-by- $(\sigma_1(\alpha), \dots, \sigma_n(\alpha))$ is just the diagonal matrix with $\sigma_i(\alpha)$ in the i th diagonal entry. Similarly,

$$T_{L/K}(\alpha) = T_{(L \otimes_K \Omega)/\Omega}(\alpha \otimes 1) = T_{\Omega^n/\Omega}(\sigma_1(\alpha), \dots, \sigma_n(\alpha)) = \sum_{i=1}^n \sigma_i(\alpha). \quad \square$$

The proof above demonstrates a useful trick: when working over a field that is not algebraically/separably closed, base change to an algebraic/separable closure. This often turns separable field extensions into étale algebras that are no longer fields.

Proposition 5.6. Let L/K be a (not necessarily separable) finite extension, let $\bar{K}$ be an algebraic closure of K containing L . Let $\alpha \in L^\times$ have minimal polynomial $f \in K[x]$ with factorization $f(x) = \prod_{i=1}^d (x - \alpha_i)$ in $\bar{K}[x]$, and let $e = [L : K(\alpha)]$. We have

$$N_{L/K}(\alpha) = \prod_{i=1}^d \alpha_i^e \quad \text{and} \quad T_{L/K}(\alpha) = e \sum_{i=1}^d \alpha_i.$$

In particular, if $f(x) = \sum_{i=0}^d a_i x^i$, then $N_{L/K}(\alpha) = (-1)^{de} a_0^e$ and $T_{L/K}(\alpha) = -e a_{d-1}$.

Proof. See Problem Set 2. $\square$

Corollary 5.7. Let A be an integrally closed domain with fraction field K and let L/K be a finite extension. If $\alpha \in L$ is integral over A then $N_{L/K}(\alpha) \in A$ and $T_{L/K}(\alpha) \in A$.

Proof. This follows immediately from Propositions 1.28 and 5.6. $\square$

Theorem 5.8 (TRANSITIVITY OF NORM AND TRACE). *Let $A \subseteq B \subseteq C$ be rings with C free of finite rank over B and B free of finite rank over A . Then C is free of finite rank over A and*

$$N_{C/A} = N_{B/A} \circ N_{C/B} \quad \text{and} \quad T_{C/A} = T_{B/A} \circ T_{C/B}.$$

Proof. See [3, §III.9.4]. $\square$

5.2 Dual modules, pairings, and lattices

In this section we work in a more general setting, where A is any commutative (unital) ring.

Definition 5.9. Let A be a commutative ring and M an A -module. The *dual module* $M^\vee$ is the A -module $\text{Hom}_A(M, A)$ with scalar multiplication $(af)(m) = af(m)$, where $a \in A$, $f \in \text{Hom}_A(M, A)$, and $m \in M$. If $\varphi: M \rightarrow N$ is an A -module homomorphism, the dual homomorphism $\varphi^\vee: N^\vee \rightarrow M^\vee$ is defined by $\varphi^\vee(g)(m) = g(\varphi(m))$, for $g \in N^\vee$ and $m \in M$.

It is easy to check that taking duals preserves identity maps and is compatible with composition: if $\varphi_1: M \rightarrow N$ and $\varphi_2: N \rightarrow P$ are A -module homomorphisms, then $(\varphi_2\varphi_1)^\vee = \varphi_1^\vee\varphi_2^\vee$. We thus have a contravariant functor from the category of A -modules to itself. This functor is compatible with (finite) direct sums, $(M \oplus N)^\vee \simeq M^\vee \oplus N^\vee$.

Lemma 5.10. *Let A be a commutative ring. For all A -modules M and N the A -modules $(M \oplus N)^\vee$ and $M^\vee \oplus N^\vee$ are canonically isomorphic.*

Proof. We have inverse A -module homomorphisms $\varphi \mapsto (m \mapsto \varphi(m, 0), n \mapsto \varphi(0, n))$ and $(\phi, \psi) \mapsto ((m, n) \mapsto \phi(m) + \psi(n))$. $\square$

If A is a field and M is finitely generated, then M is a vector space of finite dimension, $M^\vee$ is its dual space and we have $M^{\vee\vee} \simeq M$. In general not every A -module is isomorphic to its double dual; those that are are said to be *reflexive*.

We have already seen examples of reflexive modules: every invertible fractional ideal is isomorphic to the dual of its inverse, hence to its double dual, and is thus reflexive.

Proposition 5.11. *Let A be an integral domain with fraction field K and let M be a nonzero A -submodule of K . Then $M^\vee \simeq (A : M) := \{x \in K : xM \subseteq A\}$; in particular, if M is an invertible fractional ideal then $M^\vee \simeq M^{-1}$ and $M^{\vee\vee} \simeq M$.*

Proof. For any $x \in (A : M)$ the map $m \mapsto xm$ is an A -linear map from M to A , hence an element of $M^\vee$, and this defines an A -module homomorphism $\varphi: (A : M) \rightarrow M^\vee$, since the map $x \mapsto (m \mapsto xm)$ is itself A -linear. Since $M \subseteq K$ is a nonzero A -module, it contains some nonzero $a \in A$ (if $a/b \in M$, so is $ba/b = a$). If $f \in M^\vee$ and $m = b/c \in M$ then

$$f(m) = f\left(\frac{b}{c}\right) = \frac{ac}{ac}f\left(\frac{b}{c}\right) = \frac{b}{ac}f\left(\frac{ac}{c}\right) = \frac{b}{ac}f(a) = \frac{f(a)}{a}m,$$

where we have used the fact that $a_1f(a_2/a_3) = a_2f(a_1/a_3)$ for any $a_1, a_2, a_3 \in A$ with $a_1/a_3, a_2/a_3 \in M$, by the A -linearity of f . It follows that f corresponds to multiplication by $x = f(a)/a$, which lies in $(A : M)$ since $xm = f(m) \in A$ for all $m \in M$. The map $f \mapsto f(a)/a$ defines an A -module homomorphism $M^\vee \rightarrow (A : M)$ inverse to φ , so φ is an isomorphism. When M is an invertible fractional ideal we have $M^\vee \simeq (A : M) = M^{-1}$, by Lemma 2.20, and $M^{\vee\vee} \simeq (M^{-1})^{-1} = M$ follows. $\square$

Example 5.12. As a $\mathbb{Z}$ -module, we have $\mathbb{Q}^\vee = \{0\}$ because there are no non-trivial $\mathbb{Z}$ -linear homomorphisms from $\mathbb{Q}$ to $\mathbb{Z}$; indeed, $\mathbb{Q}$ is a divisible group and $\mathbb{Z}$ contains no non-trivial divisible subgroups. It follows that $\mathbb{Q}^{\vee\vee} = \{0\}$ (but as $\mathbb{Q}$ -modules we have $\mathbb{Q} \simeq \mathbb{Q}^\vee \simeq \mathbb{Q}^{\vee\vee}$). Similarly, the dual of any finite $\mathbb{Z}$ -module (any finite abelian group) is the zero module, as is the double dual. More generally, if A is an integral domain every dual (and double dual) A -module must be torsion free, but not all A -modules are torsion free.

One situation where we can recover many of the standard results that hold for vector spaces of finite dimension (with essentially the same proofs), is when M is a free module of finite rank. In particular, not only is M reflexive, we have $M \simeq M^\vee$ (non-canonically) and may explicitly construct a dual basis.

Theorem 5.13. *Let A be a commutative ring and let M be a free A -module of rank n . Then $M^\vee$ is also a free A -module of rank n , and each basis $(e_1, \dots, e_n)$ of M uniquely determines a dual basis $(e_1^\vee, \dots, e_n^\vee)$ of $M^\vee$ with the property*

$$e_i^\vee(e_j) = \delta_{ij} := \begin{cases} 1 & i = j, \\ 0 & i \neq j. \end{cases}$$

Proof. If $n = 0$ then $M = M^\vee = \{0\}$ and the theorem holds. Now assume $n \geq 1$ and fix an A -basis $\mathbf{e} := (e_1, \dots, e_n)$ for M . For each $\mathbf{a} := (a_1, \dots, a_n) \in A^n$, define $f_{\mathbf{a}} \in M^\vee$ by setting $f_{\mathbf{a}}(e_i) = a_i$ and extending A -linearly. The map $\mathbf{a} \mapsto f_{\mathbf{a}}$ gives an A -module homomorphism $A^n \rightarrow M^\vee$ with inverse $f \mapsto (f(e_1), \dots, f(e_n))$ and is therefore an isomorphism. It follows that $M^\vee \simeq A^n$ is a free A -module of rank n .

Now let $e_i^\vee := f_{\hat{\mathbf{e}}_i}$, where $\hat{\mathbf{e}}_i := (0, \dots, 0, 1, 0, \dots, 0) \in A^n$ has a 1 in the i th position. Then $\mathbf{e}^\vee := (e_1^\vee, \dots, e_n^\vee)$ is a basis for $M^\vee$, since $(\hat{\mathbf{e}}_1, \dots, \hat{\mathbf{e}}_n)$ is a basis for A^n , and $e_i^\vee(e_j) = \delta_{ij}$. The basis $\mathbf{e}^\vee$ is uniquely determined by $\mathbf{e}$: it must be the image of $(\hat{\mathbf{e}}_1, \dots, \hat{\mathbf{e}}_n)$ under the isomorphism $\mathbf{a} \mapsto f_{\mathbf{a}}$ determined by $\mathbf{e}$. $\square$

Definition 5.14. Let A be a commutative ring and M an A -module. A (bilinear) *pairing* on M is an A -linear map $\langle \cdot, \cdot \rangle: M \times M \rightarrow A$. Explicitly, this means that for all $u, v, w \in M$ and $\lambda \in A$ we have

$$\begin{aligned} \langle u + v, w \rangle &= \langle u, w \rangle + \langle v, w \rangle, \\ \langle u, v + w \rangle &= \langle u, v \rangle + \langle u, w \rangle, \\ \langle \lambda u, v \rangle &= \langle u, \lambda v \rangle = \lambda \langle u, v \rangle. \end{aligned}$$

If $\langle v, w \rangle = \langle w, v \rangle$ then $\langle \cdot, \cdot \rangle$ is *symmetric*, if $\langle v, w \rangle = -\langle w, v \rangle$ then $\langle \cdot, \cdot \rangle$ is *skew-symmetric*, and if $\langle v, v \rangle = 0$ then $\langle \cdot, \cdot \rangle$ is *alternating* (the last two are equivalent provided $\text{char}(A) \neq 2$). The pairing $\langle \cdot, \cdot \rangle$ induces an A -module homomorphism

$$\begin{aligned} \varphi: M &\rightarrow M^\vee \\ m &\mapsto (n \mapsto \langle m, n \rangle) \end{aligned}$$

If $\ker \varphi = \{0\}$ then $\langle \cdot, \cdot \rangle$ is *nondegenerate*, and if φ is an isomorphism then $\langle \cdot, \cdot \rangle$ is *perfect*.

Every perfect pairing is necessarily nondegenerate. If M is a vector space of finite dimension the converse holds, but this is not true in general, not even for free modules of finite rank: consider the pairing $\langle x, y \rangle := 2xy$ on $\mathbb{Z}$, which is nondegenerate but not perfect.

If M is a free A -module with basis $(e_1, \dots, e_n)$ and $\langle \cdot, \cdot \rangle$ is a perfect pairing, we can apply the inverse of the isomorphism $\varphi: M \xrightarrow{\sim} M^\vee$ induced by the pairing to the dual basis $(e_1^\vee, \dots, e_n^\vee)$ given by Theorem 5.13 to obtain a basis $(e'_1, \dots, e'_n)$ for M that satisfies

$$\langle e'_i, e_j \rangle = \delta_{ij}.$$

When $\langle \cdot, \cdot \rangle$ is symmetric we can similarly recover $(e_1, \dots, e_n)$ from $(e'_1, \dots, e'_n)$ in the same way. We record this fact in the following proposition.

Proposition 5.15. *Let A be a commutative ring and let M be a free A -module of rank n with a perfect pairing $\langle \cdot, \cdot \rangle$. For each A -basis $(e_1, \dots, e_n)$ of M there is a unique basis $(e'_1, \dots, e'_n)$ for M such that $\langle e'_i, e_j \rangle = \delta_{ij}$.*

Proof. Existence follows from the discussion above: apply the inverse of the isomorphism $\varphi: V \rightarrow V^\vee$ induced by $\langle \cdot, \cdot \rangle$ to the dual basis $(e_1^\vee, \dots, e_n^\vee)$ given by Theorem 5.13 to obtain a basis $(e'_1, \dots, e'_n)$ for M with $e'_i = \varphi^{-1}(e_i^\vee)$. We then have $e_i^\vee = \varphi(e'_i) = m \mapsto \langle e'_i, m \rangle$ and

$$\langle e'_i, e_j \rangle = \varphi(e'_i)(e_j) = e_i^\vee(e_j) = \delta_{ij}$$

for $1 \leq i, j \leq n$. If $(f'_1, \dots, f'_n)$ is another basis for M with the same property then for each i we have $\langle e'_i - f'_i, e_j \rangle = \delta_{ij} - \delta_{ij} = 0$ for every e_j , and therefore $\langle e'_i - f'_i, m \rangle = 0$ for all $m \in M$, but then $e'_i - f'_i \in \ker \varphi = \{0\}$, since the perfect pairing $\langle \cdot, \cdot \rangle$ is nondegenerate, and therefore $f'_i = e'_i$ for each i ; uniqueness follows. $\square$

Remark 5.16. In what follows the commutative ring A in Proposition 5.15 will typically be a field K and the free A -module M will be a K -vector space that we will denote V . We may then use A to denote a subring of K and M to denote an A -submodule of V . A perfect pairing $\langle \cdot, \cdot \rangle$ on the K -vector space V will typically not restrict to a perfect pairing on the A -module M . For example, the perfect pairing $\langle x, y \rangle = xy$ on $\mathbb{Q}$ does not restrict to a perfect pairing on the $\mathbb{Z}$ -module $2\mathbb{Z}$ because the induced map $\varphi: 2\mathbb{Z} \rightarrow 2\mathbb{Z}^\vee$ defined by $\varphi(m) = (n \mapsto mn)$ is not surjective: the map $x \mapsto x/2$ lies in $2\mathbb{Z}^\vee = \text{Hom}_{\mathbb{Z}}(2\mathbb{Z}, \mathbb{Z})$ but it is not in the image of φ .

We now introduce the notion of a lattice in a vector space.

Definition 5.17. Let A be an integral domain with fraction field K and let V be a K -vector space of finite dimension. A (full) A -lattice in V is a finitely generated A -submodule M of V that spans V as a K -vector space.

Remark 5.18. Some authors require A -lattices to be free A -modules. When $A = \mathbb{Z}$ (or any PID) this is not a restriction because M is necessarily torsion-free (it lies in a vector space) and any finitely generated torsion-free module over a PID is free (by the structure theorem for finitely generated modules over a PID). But when A is not a PID, finitely generated torsion-free A -modules will typically *not be free*. We do not want to exclude this case! In particular if L/K is an extension of number fields the ring of integers $\mathcal{O}_L$ will typically not be a free $\mathcal{O}_K$ -module (even though it is a free $\mathbb{Z}$ -module, as we shall shortly prove), but we still want to treat $\mathcal{O}_L$ as an $\mathcal{O}_K$ -lattice in L (this will be important in later lectures when we define the *different ideal* $\mathcal{D}_{L/K}$).

Definition 5.19. Let A be a noetherian domain with fraction field K , and let V be a K -vector space of finite dimension with a perfect pairing $\langle \cdot, \cdot \rangle$. If M is an A -lattice in V , its *dual lattice* (with respect to the perfect pairing $\langle \cdot, \cdot \rangle$ on V) is the A -module

$$M^* := \{x \in V : \langle x, m \rangle \in A \text{ for all } m \in M\}.$$

It is clear that M^* is an A -submodule of V , but it is not clear that it is an A -lattice in V (it must be finitely generated and span V), nor is it obvious that it is isomorphic to the dual module $M^\vee$. In order to justify the term *dual lattice*, let us now prove both facts. We will need to use the hypothesis that A is noetherian, since in general the dual of a finitely generated A -module need not be finitely generated. Notice that $\langle \cdot, \cdot \rangle$ is a perfect pairing on the K -module V that need not restrict to a perfect pairing on the A -module M .

Theorem 5.20. *Let A be a noetherian domain with fraction field K , let V be a K -vector space with a perfect pairing $\langle \cdot, \cdot \rangle$, and let M be an A -lattice in V . The dual lattice M^* is an A -lattice in V isomorphic to $M^\vee$.*

Proof. Let $\mathbf{e} := (e_1, \dots, e_n)$ be a K -basis for V that lies in M , and let $\mathbf{e}' := (e'_1, \dots, e'_n)$ be the unique K -basis for V given by Proposition 5.15 that satisfies $\langle e'_i, e_j \rangle = \delta_{ij}$.

To show that M^* spans V we write a finite set S of generators for M in terms of the basis $\mathbf{e}$ with coefficients in K and let d be the product of all denominators that appear. We claim that $d\mathbf{e}'$ lies in M^* : for each e'_i and generator $m \in S$, if we put $m = \sum_j m_j e_j$ then

$$\langle de'_i, m \rangle = d \langle e'_i, \sum_j m_j e_j \rangle = d \sum_j m_j \langle e'_i, e_j \rangle = d \sum_j m_j \delta_{ij} = dm_i \in A,$$

by our choice of d , and this implies $de'_i \in M^*$. Thus M^* contains a basis $d\mathbf{e}'$ for V .

We now show M^* is finitely generated. Let

$$N := \{a_1 e_1 + \dots + a_n e_n : a_1, \dots, a_n \in A\} \simeq A^n$$

be the free A -submodule of M spanned by $\mathbf{e}$. The A -module N contains a basis for V and is finitely generated, so it is an A -lattice in V . The K -basis $\mathbf{e}'$ for V lies in N^* , since $\langle e'_i, e_j \rangle = \delta_{ij} \in A$, and we claim it is an A -basis for N^* . Given $x \in N^*$, if we write $x = \sum_i x_i e'_i$ then $\langle x, e_i \rangle = x_i \langle e'_i, e_i \rangle = x_i$ lies in A , since $x \in N^*$, so x lies in the A -span of $\mathbf{e}'$. It follows that N^* is a free A -module of rank n , and in particular, a finitely generated module over a noetherian ring and therefore a noetherian module (a module whose submodules are all finitely generated); see [1, Thm. 16.19]. From the definition of the dual lattice we have $N \subseteq M \Rightarrow M^* \subseteq N^*$, so M^* is a submodule of a noetherian module, hence finitely generated.

We now show $M^* \simeq M^\vee$. We have an obvious A -module homomorphism $\varphi: M^* \rightarrow M^\vee$ given by $x \mapsto (m \mapsto \langle x, m \rangle)$, and the A -module homomorphism $\psi: M^\vee \rightarrow M^*$ defined by $f \mapsto \sum_i f(e_i) e'_i$ is the inverse of φ . Indeed, for any $x = \sum_i x_i e'_i \in M^*$ we have

$$\psi(\varphi(x)) = \sum_i \varphi(x)(e_i) e'_i = \sum_i \langle x, e_i \rangle e'_i = \sum_i \sum_j x_j \langle e'_j, e_i \rangle e'_i = \sum_i x_i e'_i = x,$$

and for any $f \in M^\vee$ and each generator $m = \sum_j m_j e_j$ for M we have

$$\varphi(\psi(f))(m) = \varphi(\sum_i f(e_i) e'_i)(m) = \sum_i \varphi(f(e_i) e'_i)(m) = \sum_i \langle f(e_i) e'_i, \sum_j m_j e_j \rangle = f(m),$$

which implies $\varphi(\psi(f)) = f$ and $\varphi^{-1} = \psi$; thus φ is an isomorphism from M^* to $M^\vee$. $\square$

Corollary 5.21. *Let A be a noetherian domain with fraction field K . If M_1, M_2 are A -lattices in K -vector spaces V_1, V_2 with perfect pairings $\langle \cdot, \cdot \rangle_1, \langle \cdot, \cdot \rangle_2$ (resp.), then $\langle \cdot, \cdot \rangle_1 + \langle \cdot, \cdot \rangle_2$ defines a perfect pairing on $V_1 \oplus V_2$ and $(M \oplus N)^* \simeq M^* \oplus N^*$.*

Proof. This follows from Lemma 5.10 and Theorem 5.20. $\square$

Corollary 5.22. *Let A be a noetherian domain with fraction field K , let V be a K -vector space with a perfect pairing $\langle \cdot, \cdot \rangle$, and let M be a free A -lattice in V with A -basis $(e_1, \dots, e_n)$. The dual lattice M^* is a free A -lattice in V that has a unique A -basis $(e_1^*, \dots, e_n^*)$ that satisfies $\langle e_i^*, e_j \rangle = \delta_{ij}$.*

Proof. This follows from the proof of Theorem 5.20 with $N = M$ and $e_i^* = e'_i$. $\square$

You might wonder whether $M^{**} = M$ for an A -lattice M in a vector space V . This is false in general, but it is true when A is a Dedekind domain and we have a symmetric perfect pairing on V . To prove this we first show that the dual lattice respects localization.

Lemma 5.23. *Let A be a noetherian domain with fraction field K , let V be a K -vector space of finite dimension with a perfect pairing $\langle \cdot, \cdot \rangle$, let M be an A -lattice in V , and let S be a multiplicative subset of A . Then $S^{-1}M$ and $S^{-1}M^*$ are $(S^{-1}A)$ -lattices in V satisfying $(S^{-1}M)^* = S^{-1}M^*$.*

Proof. It is clear that $S^{-1}M$ and $S^{-1}M^*$ are both $S^{-1}A$ -lattices: each contains a basis for V (since M and M^* do), and both are finitely generated as $S^{-1}A$ -modules (since M and M^* are finitely generated as A -modules).

Let $m_1, \dots, m_n$ be A -module generators for M (and therefore $S^{-1}A$ -module generators for $S^{-1}M$). If x is an element of $(S^{-1}M)^*$ then for each m_i we have $\langle x, m_i \rangle = a_i/s_i$ for some $a_i \in A$ and $s_i \in S$, and if we put $s = s_1 \cdots s_n$ then $\langle sx, m_i \rangle \in A$ for every m_i , hence for all $m \in M$; thus $sx \in M^*$ and $x \in S^{-1}M^*$. Conversely, if $x = y/s$ is an element of $S^{-1}M^*$ with $y \in M^*$ and $s \in S$, then $\langle y, m_i \rangle \in A$ for every m_i and $\langle x, m_i \rangle = \langle y, m_i \rangle/s \in S^{-1}A$ for every m_i , hence for all $m \in S^{-1}M$, and it follows that $x \in (S^{-1}M)^*$. $\square$

Proposition 5.24. *Let A be a Dedekind domain with fraction field K , let V be a K -vector space of finite dimension with a symmetric perfect pairing $\langle \cdot, \cdot \rangle$, and let M be an A -lattice in V . Then $M^{**} = M$.*

Proof. By Proposition 2.6, it suffices to show $(M^{**})_{\mathfrak{p}} = M_{\mathfrak{p}}$ for each maximal ideal $\mathfrak{p}$ of A . By Lemma 5.23 we have $(M^{**})_{\mathfrak{p}} = M_{\mathfrak{p}}^{**}$, so it is enough to show that the proposition holds when A is replaced by one of its localizations $A_{\mathfrak{p}}$ (a DVR, since A is a Dedekind domain).

So let us assume that A is a DVR. Then A is a PID and M and M^* are both torsion-free modules over a PID, hence free A -modules. So let us choose an A -basis $(e_1, \dots, e_n)$ for M , and let $(e_1^*, \dots, e_n^*)$ be the unique dual A -basis for M^* that satisfies $\langle e_i^*, e_j \rangle = \delta_{ij}$ (given by Corollary 5.22). If we now let $(e_1^{**}, \dots, e_n^{**})$ be the unique A -basis for M^{**} that satisfies $\langle e_i^{**}, e_j^* \rangle = \delta_{ij}$ and note that $\langle e_i, e_j^* \rangle = \delta_{ij}$ (since $\langle \cdot, \cdot \rangle$ is symmetric), by uniqueness, we must have $e_i^{**} = e_i$ for all i , and therefore $M^{**} = M$. $\square$

5.3 Extensions of Dedekind domains

Let A be a Dedekind domain with fraction field K , let L/K be a finite extension, and let B be the integral closure of A in L . We wish to prove that B is a Dedekind domain, which we will do by showing that it is an A -lattice in L ; this will imply, in particular, that B is finitely generated, which is really the only difficult thing to show. Let us first show that B spans L as a vector space (and in fact L is its fraction field).

Proposition 5.25. *Let A be a Dedekind domain with fraction field K , let L/K be a finite extension, and let B be the integral closure of A in L . Every element of L can be written as b/a with $a \in A$ and $b \in B$. In particular, B spans L as a K -vector space and L is the fraction field of B .*

Proof. Let $\alpha \in L$. By multiplying the minimal polynomial of α in $K[x]$ by the product of the denominators of its coefficients, we obtain a polynomial in $A[x]$:

$$g(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0,$$

with $a_n \neq 0$, that has α as a root. We can make this polynomial monic by replacing x with x/a_n and multiplying through by a_n^{n-1} to obtain

$$a_n^{n-1} g(x/a_n) = x^n + a_{n-1} x^{n-1} + a_n a_{n-2} x^{n-2} \cdots + a_n^{n-2} a_1 x + a_n^{n-1} a_0.$$

This is a monic polynomial with coefficients in A that has $a_n \alpha \in L$ as a root. Therefore $a_n \alpha \in B$, since B is the integral closure of A in L , and $\alpha = b/a_n$ for some $b \in B$ and $a_n \in A$ as claimed. It follows that B generates L as a K -vector space (we have $\alpha = b \cdot \frac{1}{a_n}$ with $\frac{1}{a_n} \in K$), and $B \subseteq L \subseteq \text{Frac } B$ implies $L = \text{Frac } B$ (no smaller field can contain B). $\square$

Proposition 5.26. *Let A be a Dedekind domain with fraction field K , let L/K be a finite extension of fields, and let B be the integral closure of A in L . Then $N_{L/K}(b) \in A$ and $T_{L/K}(b) \in A$ for all $b \in B$.*

Proof. The minimal polynomial $f = \sum_{i=0}^d a_i x^i \in K[x]$ of b has coefficients in A , by Proposition 1.28, and it then follows from Proposition 5.6 that $N_{L/K}(b) = (-1)^{de} a_0^e \in A$ and $T_{L/K}(b) = -e a_{d-1} \in A$ (where $e = [L : K(b)] \in \mathbb{Z}$). $\square$

Definition 5.27. Let B/A be a ring extension with B a free A -module of finite rank. The *trace pairing* on B is the map $B \times B \rightarrow A$ defined by

$$\langle x, y \rangle_{B/A} := T_{B/A}(xy).$$

Theorem 5.28. *Let L be a commutative K -algebra of finite dimension. The trace pairing $\langle \cdot, \cdot \rangle_{L/K}$ is a symmetric bilinear pairing. It is a perfect pairing if and only if L is a finite étale K -algebra.*

Proof. Bilinearity follows from the K -linearity of the trace map $T_{L/K}$, and symmetry is immediate. The fact that L is a K -vector space implies that the trace pairing is perfect if and only if it is nondegenerate.

If L is not reduced then the proposition holds, since it is not étale (by Theorem 4.40), and the trace pairing is degenerate: for any nonzero nilpotent x the map $y \mapsto T_{L/K}(xy)$ must be the zero map, since every xy is also nilpotent and the trace of any nilpotent element z is zero (the matrix of the multiplication-by- z map is nilpotent, so its trace is zero).

We now assume L is reduced, hence semisimple (by Lemma 4.41) and thus a product of fields. It suffices to consider the case that L is a field, since the trace pairing on a product of field extensions is nondegenerate if and only if the trace pairing on each factor is nondegenerate, and a product of field extensions is étale if and only if each factor is étale.

As proved on Problem Set 2, $T_{L/K}$ is the zero map if and only if the field extension L/K is inseparable. If $T_{L/K}$ is the zero map then the trace pairing is clearly degenerate, and otherwise we may pick $z \in L$ for which $T_{L/K}(z) \neq 0$. Then for every $x \in L^\times$ we have $\langle x, z/x \rangle_{L/K} = T_{L/K}(z) \neq 0$, so $x \mapsto \langle x, y \rangle_{L/K}$ is not the zero map, and it follows that the trace pairing is nondegenerate. $\square$

Remark 5.29. Theorem 5.28 gives another equivalent definition of a finite étale K -algebra in addition to the six listed in Theorem 4.40: a finite étale K -algebra is a commutative K -algebra of finite dimension for which the trace pairing is a perfect pairing.

We now assume that L/K is separable. For the next several lectures we will be working in the following setting: A is a Dedekind domain with fraction field K , the extension L/K is finite separable, and B is the integral closure of A in L (which we will shortly prove is a Dedekind domain). As a convenient shorthand, we will write “assume $AKLB$ ” to indicate that we are using this setup.

Proposition 5.30. *Assume $AKLB$. Then B is an A -lattice in L , and in particular, B is finitely generated as an A -module.*

Proof. By Proposition 5.25, B spans L as a K -vector space, so it contains a basis $(e_1, \dots, e_n)$ for L as a K -vector space. Let $M \subseteq B$ be the A -span of $(e_1, \dots, e_n)$. Then M is an A -lattice in L contained in B , and it has a dual lattice M^* that contains the A -module

$$B^* := \{x \in L : \langle x, b \rangle_{L/K} \in A \text{ for all } b \in B\}.$$

Proposition 5.26 implies that $B \subseteq B^*$, and we thus have inclusions

$$M \subseteq B \subseteq B^* \subseteq M^*.$$

By Theorem 5.20, M^* is an A -lattice in L , hence finitely generated, hence noetherian. It follows that its A -submodule B is finitely generated and thus an A -lattice in L . $\square$

Remark 5.31. When L/K is inseparable, B need not be finitely generated as an A -module, not even when A is a PID; see [2, Ex. 11, p. 205]. We used the separability hypothesis in order to get a perfect pairing, which plays a crucial role in the proof of Theorem 5.20.

Lemma 5.32. *Let B/A be an extension of domains with B integral over A , and let $\mathfrak{q}_0 \subsetneq \mathfrak{q}_1$ be primes of B . Then $\mathfrak{q}_0 \cap A \subsetneq \mathfrak{q}_1 \cap A$ and $\dim A \geq \dim B$.*

Proof. We first replace B with $B/\mathfrak{q}_0$ and replace A , $\mathfrak{q}_0$, and $\mathfrak{q}_1$ with their images in $B/\mathfrak{q}_0$ (the new B is integral over the new A , since the image of a monic polynomial in $A[x]$ is a monic polynomial in $(A/(\mathfrak{q}_0 \cap A))[x]$). Then $\mathfrak{q}_0 = (0)$ and $\mathfrak{q}_1$ is a nonzero prime ideal. Let $\alpha \in \mathfrak{q}_1$ be nonzero. Its minimal polynomial $x^n + a_{n-1}x^{n-1} + \dots + a_0$ over K has coefficients in A (since $\alpha \in \mathfrak{q}_1 \subseteq B$ is integral over A), with $a_0 \neq 0$ (otherwise divide by x). We have $a_0 = -a_1\alpha - \dots - \alpha^n \in \mathfrak{q}_1$, thus $0 \neq a_0 \in \mathfrak{q}_1 \cap A$. So $\mathfrak{q}_1 \cap A$ is not the zero ideal and therefore properly contains $\mathfrak{q}_0 \cap A = \{0\}$. We can apply this result repeatedly to any chain of distinct prime ideals in B to get a corresponding chain of distinct prime ideals in A . It follows that $\dim A \geq \dim B$. $\square$

Theorem 5.33. *Let A be a Dedekind domain with fraction field K , let L/K be a finite separable extension, and let B be the integral closure of A in L . Then B is a Dedekind domain.*

Proof. Recall that we defined a Dedekind domain as an integrally closed noetherian domain of dimension at most one. Let us verify that each of these conditions holds:

- B is an integrally closed domain (by definition);
- B is finitely generated over the noetherian ring A (by Prop. 5.30), hence noetherian;
- B has dimension at most 1, since $\dim B \leq \dim A \leq 1$, by Lemma 5.32.

Thus B is a Dedekind domain. $\square$

Remark 5.34. Theorem 5.33 holds without the assumption that L/K is separable. This follows from the Krull-Akizuki Theorem, see [5, Thm. 11.7] or [4, §VII.2.5], which is used to prove that B is noetherian even when it is not finitely generated as an A -module.

Corollary 5.35. *The ring of integers of a number field is a Dedekind domain.*

5.4 Splitting primes in Dedekind extensions

We continue in the $AKLB$ setup, in which A is a Dedekind domain, K is its fraction field, L/K is a finite separable¹ extension, and B is the integral closure of A , which we now know is a Dedekind domain with fraction field L . As we proved in earlier lectures, every nonzero ideal in a Dedekind domain can be uniquely factored into prime ideals. Understanding the ideal structure of a Dedekind domain thus boils down to understanding its prime ideals. In order to simplify the language, whenever we have a Dedekind domain A , by a *prime* of A (or of its fraction field K), we always mean a **nonzero prime ideal** of A .

If A has dimension zero then so does B , in which case there are no primes to consider, so we may as well assume $\dim A = 1$, in which case $\dim B = 1$ as well (if B is a field then so is $B \cap K = A$). Henceforth our $AKLB$ setup will include the assumption that $A \neq K$.

Given a prime $\mathfrak{p}$ of A , we can consider the ideal $\mathfrak{p}B$ it generates in B (its extension to B under the inclusion map). The ideal $\mathfrak{p}B$ need not be prime, but it can be uniquely factored into nonzero prime ideals in the Dedekind domain B . We thus have

$$\mathfrak{p}B = \prod_{\mathfrak{q}} \mathfrak{q}^{e_{\mathfrak{q}}},$$

where $\mathfrak{q}$ ranges over primes of B and the exponents $e_{\mathfrak{q}} \geq 0$ are zero for all but finitely many primes $\mathfrak{q}$. The primes $\mathfrak{q}$ for which $e_{\mathfrak{q}} > 0$ are said to *lie over* or *above* the prime ideal $\mathfrak{p}$. As an abuse of notation, we will often write $\mathfrak{q}|\mathfrak{p}$ to indicate this relationship (there is little risk of confusion, the prime ideal $\mathfrak{p}$ is maximal hence not divisible by any prime ideals of A other than itself).

Lemma 5.36. *Let A be a ring of dimension one contained in a Dedekind domain B . Let $\mathfrak{p}$ be a prime of A and let $\mathfrak{q}$ be a prime of B . Then $\mathfrak{q}|\mathfrak{p}$ if and only if $\mathfrak{q} \cap A = \mathfrak{p}$.*

Proof. If $\mathfrak{q}$ divides $\mathfrak{p}B$ then it contains $\mathfrak{p}B$ (to divide is to contain), and therefore $\mathfrak{q} \cap A$ contains $\mathfrak{p}B \cap A$ which contains $\mathfrak{p}$; the ideal $\mathfrak{p}$ is maximal and $\mathfrak{q} \cap A \neq A$ (since $1 \notin \mathfrak{q}$), so $\mathfrak{q} \cap A = \mathfrak{p}$. Conversely, if $\mathfrak{q} \cap A = \mathfrak{p}$ then $\mathfrak{q} = \mathfrak{q}B$ certainly contains $(\mathfrak{q} \cap A)B = \mathfrak{p}B$, and B is a Dedekind domain, so $\mathfrak{q}$ divides $\mathfrak{p}B$ (in a Dedekind domain to contain is to divide). $\square$

Lemma 5.36 implies that contraction gives us a surjective map $\text{Spec } B \rightarrow \text{Spec } A$ defined by $\mathfrak{q} \mapsto \mathfrak{q} \cap A$; to see why it is surjective, note that $(0) \cap A = (0)$, and if $\mathfrak{p}$ is a nonzero element of $\text{Spec } A$ then $\mathfrak{p}B$ is nonzero and not the unit ideal, and therefore divisible by at least one $\mathfrak{q} \in \text{Spec } B$. The fibers of this map are finite; we use $\{\mathfrak{q}|\mathfrak{p}\}$ to denote the fiber above a prime $\mathfrak{p}$ of A .

The primes $\mathfrak{p}$ of A are all maximal ideals (since $\dim A = 1$), so each has an associated residue field $A/\mathfrak{p}$, and similarly for primes $\mathfrak{q}$ of B . If $\mathfrak{q}$ lies above $\mathfrak{p}$ then we may regard the residue field $B/\mathfrak{q}$ as a field extension of $A/\mathfrak{p}$: the kernel of the map $A \hookrightarrow B \rightarrow B/\mathfrak{q}$ is $\mathfrak{p} = A \cap \mathfrak{q}$, and the induced map $A/\mathfrak{p} = A/(\mathfrak{q} \cap A) \rightarrow B/\mathfrak{q}$ is a ring homomorphism of fields, hence injective.

Definition 5.37. Assume $AKLB$, and let $\mathfrak{p}$ be a prime of A . The exponent $e_{\mathfrak{q}}$ in the factorization $\mathfrak{p}B = \prod_{\mathfrak{q}|\mathfrak{p}} \mathfrak{q}^{e_{\mathfrak{q}}}$ is the *ramification index* of $\mathfrak{q}$, and the degree $f_{\mathfrak{q}} = [B/\mathfrak{q} : A/\mathfrak{p}]$

¹Most of our proofs will not actually use the separability hypothesis (and even when they do, there may be another way to prove the same result, as with Theorem 5.33). In order to simplify the presentation we will use the separability assumption whenever it would be awkward not to. The cases we are most interested in (extensions of local and global fields) are going to be separable in any event.

of the corresponding residue field extension is the *residue degree* (or *inertia degree*) of $\mathfrak{q}$. In situations where more than one extension of Dedekind domains is under consideration, we may write $e_{\mathfrak{q}/\mathfrak{p}}$ for $e_{\mathfrak{q}}$ and $f_{\mathfrak{q}/\mathfrak{p}}$ for $f_{\mathfrak{q}}$.

Lemma 5.38. *Let A be a Dedekind domain with fraction field K , let $M/L/K$ be a tower of finite separable extension, and let B and C be the integral closures of A in L and M respectively. Then C is the integral closure of B in M , and if $\mathfrak{r}$ is a prime of M lying above a prime $\mathfrak{q}$ of L lying above a prime $\mathfrak{p}$ of K then $e_{\mathfrak{r}/\mathfrak{p}} = e_{\mathfrak{r}/\mathfrak{q}}e_{\mathfrak{q}/\mathfrak{p}}$ and $f_{\mathfrak{r}/\mathfrak{p}} = f_{\mathfrak{r}/\mathfrak{q}}f_{\mathfrak{q}/\mathfrak{p}}$.*

Proof. It follows from Proposition 1.20 that the integral closure of B in M lies in C , and it contains C , since $A \subseteq B$. We thus have a tower of Dedekind extensions $C/B/A$. If $\mathfrak{r}|\mathfrak{q}|\mathfrak{p}$ then the factorization of $\mathfrak{p}C$ in C refines the factorization of $\mathfrak{p}B$ in B , so $e_{\mathfrak{r}/\mathfrak{p}} = e_{\mathfrak{r}/\mathfrak{q}}e_{\mathfrak{q}/\mathfrak{p}}$, and the residue field embedding $A/\mathfrak{p} \hookrightarrow C/\mathfrak{r}$ factors as $A/\mathfrak{p} \hookrightarrow B/\mathfrak{q} \hookrightarrow C/\mathfrak{r}$, so $f_{\mathfrak{r}/\mathfrak{p}} = f_{\mathfrak{r}/\mathfrak{q}}f_{\mathfrak{q}/\mathfrak{p}}$. $\square$

Example 5.39. Let $A := \mathbb{Z}$, with $K := \text{Frac } A = \mathbb{Q}$, and let $L := \mathbb{Q}(i)$ with $[L : K] = 2$.

The prime (5) factors in $B = \mathbb{Z}[i]$ into two distinct prime ideals:

$$5\mathbb{Z}[i] = (2+i)(2-i).$$

The prime $(2+i)$ has ramification index $e_{(2+i)} = 1$, and $e_{(2-i)} = 1$ as well. The residue field $\mathbb{Z}/(5)$ is isomorphic to the finite field $\mathbb{F}_5$, and we also have $\mathbb{Z}[i]/(2+i) \simeq \mathbb{F}_5$ (this can be determined by counting the $\mathbb{Z}[i]$ -lattice points in a fundamental parallelogram of the sublattice $(2+i)$ in $\mathbb{Z}[i]$), so $f_{(2+i)} = 1$; we similarly have $f_{(2-i)} = 1$.

The prime (7) remains prime in $B = \mathbb{Z}[i]$; its prime factorization is simply

$$7\mathbb{Z}[i] = (7),$$

where the (7) on the RHS denotes a principal ideal in B (this is clear from context). The ramification index of (7) is thus $e_{(7)} = 1$, but its residue field degree is $f_{(7)} = 2$, because $\mathbb{Z}/(7) \simeq \mathbb{F}_7$, but $\mathbb{Z}[i]/(7) \simeq \mathbb{F}_{49}$ has dimension 2 has an $\mathbb{F}_7$ -vector space.

The prime (2) factors as

$$2\mathbb{Z}[i] = (1+i)^2,$$

since $(1+i)^2 = (1+2i-1) = (2i) = (2)$ (note that i is a unit). You might be thinking that $(2) = (1+i)(1-i)$ factors into distinct primes, but note that $(1+i) = -i(1-i) = (1-i)$. Thus $e_{(1+i)} = 2$, and $f_{(1+i)} = 1$ because $\mathbb{Z}/(2) \simeq \mathbb{F}_2 \simeq \mathbb{Z}[i]/(1+i)$.

Let us now compute the sum $\sum_{\mathfrak{q}|\mathfrak{p}} e_{\mathfrak{q}}f_{\mathfrak{q}}$ for each of the primes $\mathfrak{p}$ we factored above:

$$\begin{aligned} \sum_{\mathfrak{q}|\mathfrak{p}} e_{\mathfrak{q}}f_{\mathfrak{q}} &= e_{(1+i)}f_{(1+i)} = 2 \cdot 1 = 2, \\ \sum_{\mathfrak{q}|\mathfrak{p}} e_{\mathfrak{q}}f_{\mathfrak{q}} &= e_{(2+i)}f_{(2+i)} + e_{(2-i)}f_{(2-i)} = 1 \cdot 1 + 1 \cdot 1 = 2, \\ \sum_{\mathfrak{q}|\mathfrak{p}} e_{\mathfrak{q}}f_{\mathfrak{q}} &= e_{(7)}f_{(7)} = 1 \cdot 2 = 2. \end{aligned}$$

In all three cases we obtain $2 = [\mathbb{Q}(i) : \mathbb{Q}]$; as we shall shortly prove, this is not an accident.

Example 5.40. Let $A := \mathbb{R}[x]$, with $K := \text{Frac } A = \mathbb{R}(x)$, and let $L := \mathbb{R}(\sqrt{x^3+3x})$. The integral closure of A in L is the Dedekind domain $B = \mathbb{R}[x, y]/(y^2 - x^3 - 3x)$. Then $[L : K] = 2$.

The prime $(x - 1)$ factors in B into two distinct prime ideals:

$$(x - 1) = (x - 1, y - 2)(x - 1, y + 2) \quad (\text{since } y^2 - 4 = x^3 + 3x - 4 \in (x - 1)).$$

We thus have $e_{(x-1, y-2)} = 1$, and $f_{(x-1, y-2)} = [B/(x - 1, y - 2) : A/(x - 1)] = [\mathbb{R} : \mathbb{R}] = 1$. Similarly, $e_{(x-1, y+2)} = 1$ and $f_{(x-1, y+2)} = 1$.

The prime $(x + 1)$ remains prime in B (because $y^2 = -1$ has no solutions in $\mathbb{R}$), thus $e_{(x+1)} = 1$, and $f_{(x+1)} = [B/(x + 1) : A/(x + 1)] \simeq [\mathbb{C} : \mathbb{R}] = 2$.

The prime (x) factors in B as

$$(x) = (x, y)^2,$$

and we have $e_{(x, y)} = 2$ and $f_{(x, y)} = 1$.

As in the previous example, $\sum_{\mathfrak{q}|\mathfrak{p}} e_{\mathfrak{q}} f_{\mathfrak{q}} = [L : K]$ in every case:

$$\begin{aligned} \sum_{\mathfrak{q}|(x-1)} e_{\mathfrak{q}} f_{\mathfrak{q}} &= e_{(x-1, y-2)} f_{(x-1, y-2)} + e_{(x-1, y+2)} f_{(x-1, y+2)} = 1 \cdot 1 + 1 \cdot 1 = 2, \\ \sum_{\mathfrak{q}|(x+1)} e_{\mathfrak{q}} f_{\mathfrak{q}} &= e_{(x+1)} f_{(x+1)} = 1 \cdot 2 = 2, \\ \sum_{\mathfrak{q}|(x)} e_{\mathfrak{q}} f_{\mathfrak{q}} &= e_{(x, y)} f_{(x, y)} = 2 \cdot 1 = 2, \end{aligned}$$

Before proving that $\sum_{\mathfrak{q}|\mathfrak{p}} e_{\mathfrak{q}} f_{\mathfrak{q}} = [L : K]$ always holds, let us consider the quotient ring $B/\mathfrak{p}B$. The ring $B/\mathfrak{p}B$ is typically not a field, so it is not a field extension of $A/\mathfrak{p}$, but it is an $A/\mathfrak{p}$ -algebra. This follows from the fact that B contains A and $\mathfrak{p}B$ contains $\mathfrak{p}$: given $\bar{a} \in A/\mathfrak{p}$ and $\bar{x} \in B/\mathfrak{p}B$, if we choose lifts $a \in A$ of $\bar{a}$ and $x \in B$ of $\bar{x}$ then $\bar{a}\bar{x} \in B/\mathfrak{p}B$ is the reduction of $ax \in b$ and does not depend on the choice of a and x since any other choices would be congruent modulo $\mathfrak{p}B$.

Lemma 5.41. *Assume AKLB and let $\mathfrak{p}$ be a prime of A . The dimension of $B/\mathfrak{p}B$ as an $A/\mathfrak{p}$ -vector space is equal to the dimension of L as a K -vector space.*

Proof. Let $A_{\mathfrak{p}} := S^{-1}A$ and $B_{\mathfrak{p}} := S^{-1}B$ be localizations of A and B (as A -modules), where $S = A - \mathfrak{p}$. Then $A_{\mathfrak{p}}/\mathfrak{p}A_{\mathfrak{p}} = S^{-1}A/(\mathfrak{p}S^{-1}A) \simeq A/\mathfrak{p}$ and $B_{\mathfrak{p}}/\mathfrak{p}B_{\mathfrak{p}} \simeq S^{-1}B/(\mathfrak{p}S^{-1}B) \simeq B/\mathfrak{p}B$. It follows that if the lemma is true when A is a DVR then it is true in general, so we may assume that A is a DVR, and in particular, a PID.

By Proposition 5.30, B is finitely generated as an A module, and as an integral domain containing A , it must be torsion free. It follows from the structure theorem for finitely generated modules over a PID that B is free of finite rank over A . By Proposition 5.25, B spans L as a K -vector space, so any A -basis for B is a K -basis for L . It follows that B has rank $n := [L : K]$ as a free A -module, that is, $B \simeq A^n$. We then have $\mathfrak{p}B \simeq \mathfrak{p}A^n = (\mathfrak{p}A)^n$, so $B/\mathfrak{p}B \simeq A^n/(\mathfrak{p}A)^n \simeq (A/\mathfrak{p})^n$ is a free $A/\mathfrak{p}$ -module of dimension n . $\square$

Example 5.42. Let $A = \mathbb{Z}$, $B = \mathbb{Z}[i]$, and consider $\mathfrak{p} = (2)$. We have $\mathfrak{p}B = 2\mathbb{Z}[i] = (1 + i)^2$, and $B/\mathfrak{p}B = \mathbb{Z}[i]/2\mathbb{Z}[i] = \mathbb{Z}[i]/(1 + i)^2$ is an $\mathbb{F}_2$ -algebra of dimension $2 = [\mathbb{Q}(i) : \mathbb{Q}]$. It contains a nonzero nilpotent (the image of $i + 1$), so it is not a finite étale $\mathbb{F}_2$ -algebra. It is a ring of cardinality 4 and characteristic 2 isomorphic to $\mathbb{F}_2[x]/(x^2)$.

Theorem 5.43. *Assume AKLB. For each prime $\mathfrak{p}$ of A we have*

$$\sum_{\mathfrak{q}|\mathfrak{p}} e_{\mathfrak{q}} f_{\mathfrak{q}} = [L : K].$$

Proof. We have

$$B/\mathfrak{p}B \simeq \prod_{\mathfrak{q}|\mathfrak{p}} B/\mathfrak{q}^{e_{\mathfrak{q}}}$$

Applying the previous proposition gives

$$\begin{aligned} [L : K] &= [B/\mathfrak{p}B : A/\mathfrak{p}] \\ &= \sum_{\mathfrak{q}|\mathfrak{p}} [B/\mathfrak{q}^{e_{\mathfrak{q}}} : A/\mathfrak{p}] \\ &= \sum_{\mathfrak{q}|\mathfrak{p}} e_{\mathfrak{q}} [B/\mathfrak{q} : A/\mathfrak{p}] \\ &= \sum_{\mathfrak{q}|\mathfrak{p}} e_{\mathfrak{q}} f_{\mathfrak{q}}. \end{aligned}$$

The second equality comes from the Chinese Remainder Theorem, and the third uses the fact that $B/\mathfrak{q}^{e_{\mathfrak{q}}}$ has dimension $e_{\mathfrak{q}}$ as a $B/\mathfrak{q}$ -vector space. Indeed, we have

$$\mathfrak{q}^{e_{\mathfrak{q}}} = \{x \in B : v_{\mathfrak{q}}(x) \geq e_{\mathfrak{q}}\},$$

and if $\pi \in \mathfrak{q}$ is a uniformizer for $B_{\mathfrak{q}}$ (a generator for $\mathfrak{q}B_{\mathfrak{q}}$ that we can force to lie in $\mathfrak{q}$ by clearing denominators), the images of $(\pi^0, \pi^1, \dots, \pi^{e_{\mathfrak{q}}-1})$ in $B/\mathfrak{q}^{e_{\mathfrak{q}}}$ are a $B/\mathfrak{q}$ -basis for $B/\mathfrak{q}^{e_{\mathfrak{q}}}$; to see that these elements are linearly independent, note that any non-zero $B/\mathfrak{q}$ -linear combination can be lifted to an element of B with valuation strictly less than $e_{\mathfrak{q}}$, which is therefore not an element of $\mathfrak{q}^{e_{\mathfrak{q}}}$ (hence nonzero in $B/\mathfrak{q}^{e_{\mathfrak{q}}}$). $\square$

For each prime $\mathfrak{p}$ of A , let $g_{\mathfrak{p}} := \#\{\mathfrak{q}|\mathfrak{p}\}$ denote the cardinality of the fiber above $\mathfrak{p}$.

Corollary 5.44. *Assume AKLB and let $\mathfrak{p}$ be a prime of A . Then $g_{\mathfrak{p}}$ is an integer in the interval $[1, n]$, where $n = [L : K]$, as are $e_{\mathfrak{q}}$ and $f_{\mathfrak{q}}$ for each $\mathfrak{q}|\mathfrak{p}$.*

We now define some standard terminology that we may use in the AKLB setting to describe how a prime $\mathfrak{p}$ of K splits in L (that is, for a nonzero prime ideal $\mathfrak{p}$ of A , how the ideal $\mathfrak{p}B$ factors into nonzero prime ideals $\mathfrak{q}$ of B).

Definition 5.45. Assume AKLB, let $\mathfrak{p}$ be a prime of A .

- L/K is *totally ramified* at $\mathfrak{q}$ if $e_{\mathfrak{q}} = [L : K]$ (equivalently, $f_{\mathfrak{q}} = 1 = g_{\mathfrak{p}} = 1$).
- L/K is *unramified* at $\mathfrak{q}$ if $e_{\mathfrak{q}} = 1$ **and** $B/\mathfrak{q}$ is a separable extension of $A/\mathfrak{p}$.
- L/K is *unramified above* $\mathfrak{p}$ if it is unramified at all $\mathfrak{q}|\mathfrak{p}$, equivalently, if $B/\mathfrak{p}B$ is a finite étale algebra over $A/\mathfrak{p}$.

When L/K is unramified above $\mathfrak{p}$ we say that

- $\mathfrak{p}$ *remains inert* in L if $\mathfrak{q} = \mathfrak{p}B$ is prime (equivalently, $e_{\mathfrak{q}} = g_{\mathfrak{p}} = 1$, and $f_{\mathfrak{q}} = [L : K]$).
- $\mathfrak{p}$ *splits completely* in L if $g_{\mathfrak{p}} = [L : K]$ (equivalently, $e_{\mathfrak{q}} = f_{\mathfrak{q}} = 1$ for all $\mathfrak{q}|\mathfrak{p}$).

In Example 5.42 above for the extension $\mathbb{Q}(i)/\mathbb{Q}$, the prime $\mathfrak{p} = (2)$ is ramified and the quotient ring $B/\mathfrak{p}B$ is not an étale $A/\mathfrak{p}$ algebra, even though the residue field $A/\mathfrak{p} \simeq \mathbb{F}_2$ is a perfect field (note that $B/\mathfrak{p}B$ is not a field). But when $A/\mathfrak{p}$ is a finite field (or any perfect field), for any prime $\mathfrak{q}|\mathfrak{p}$ the residue field $B/\mathfrak{q}$ is necessarily a finite étale $(A/\mathfrak{p})$ -algebra, since it must be a separable field extension, and in this case $\mathfrak{q}$ is unramified whenever $e_{\mathfrak{q}} = 1$. This applies to our primary case of interest, where L/K is an extension of global fields. However, we will occasionally want to consider Dedekind domains A whose residue fields need not be perfect, in which case $e_{\mathfrak{q}} = 1$ does not imply that $\mathfrak{q}$ is unramified.

References

- [1] Allen Altman and Steven Kleiman, *A term of commutative algebra*, Worldwide Center of Mathematics, 2013.
- [2] Zenon I. Borevich and Igor R. Shafarevich, *Number theory*, Academic Press, 1966.
- [3] Nicolas Bourbaki, *Algebra I: Chapters 1–3*, Springer, 1989.
- [4] Nicolas Bourbaki, *Commutative Algebra: Chapters 1–7*, Springer, 1989.
- [5] Hideyuki Matsumura, *Commutative ring theory*, Cambridge University Press, 1986.